Burstable MLaaS Project Kickoff

DT4D / AWS | GE Vernova Research and LTI | Internal LTI Planning

1. Project Purpose

Establish a standalone surrogate modeling environment within ARC's AWS environment, drawing on LTI's DT4D experience and the Quantum Black reference architecture. This document aligns the LTI team on the project's purpose, staffing, technical direction, and preparation for the GE kickoff.

2. Background and Objectives

GE Vernova's wind and power businesses are investing millions of dollars with Quantum Black, a McKinsey-owned firm, to develop surrogate modeling platforms. These capabilities overlap with work previously delivered through DT4D. ARC leadership sees an opportunity to rebuild that expertise internally and provide a reusable shared service across GE Vernova.

The intended offering is "surrogate in a box": an internal platform that can support wind, power, electrification, and other GE Vernova business units. The project aims to:

- Apply existing LTI DT4D knowledge and experience to accelerate delivery.
- Reduce external vendor dependency and associated costs.
- Build AWS cloud expertise within ARC and enable scalable compute capacity.
- Create a reusable surrogate modeling capability across business units.

3. Project Team and Allocation

Organization	Team Member	Planned Allocation
GE Vernova	Mike Russo; Todd Shirbak	Project stakeholders
LTI	Nam Park	100%
LTI	Chris Burke	50%
LTI	Rick Allen	50%

Nam will be dedicated full time. Chris and Rick will contribute part time, with Chris expected to provide more hours initially given his AWS architecture and HPC expertise. Rick may increase his involvement as his other project winds down. Actual weekly allocations will be coordinated as priorities evolve.

Supporting knowledge resources: Dave Garber (MLaaS Team Lead), Milton Whitney, Lalo Munoz, and Sherwin Pagtakhan. These team members are not directly assigned to delivery but may assist with environment knowledge and historical context.

4. LTI Internal Objective

Deliver Phase 1 successfully during the current contract period, which runs through February, and establish LTI as Mike Russo's trusted AWS delivery team. Strong execution should position LTI for renewal beyond the initial engagement.

5. Technical Direction

The following reflects Mike Russo's responses to the August 5 follow-up email and provides the starting point for kickoff alignment.

- **Standalone environment:** Establish a new environment within ARC using the Quantum Black design as reference architecture. The existing environment is hosted in the GE Vernova Enterprise Master Payer account; no integration with it is planned.
- **Application deployment:** Work with Jim Tallman and his engineering team to deploy their application and frontend into the new environment, then support it. Confirm the division of application, infrastructure, and support responsibilities at kickoff.
- **Available reference materials:** Some infrastructure CloudFormation templates and YAML files are available, including Karpenter scaling configuration. Application service deployment materials have not been provided.
- **Dedicated AWS account:** The account is intended solely for this application; no additional applications or services are currently planned.
- **Architecture flexibility:** The original DT4D application stack can serve as a starting point. Russo is open to alternative approaches that take advantage of subsequent technology improvements.
- **Initial technology baseline (subject to architecture review):** Kubernetes, Amazon S3, Ansys CFX, a Java Spring backend, a React frontend, and AWS compute that can scale with workload demand.

6. Kickoff Preparation and Next Steps

Action	Participants / Timing
Review DT4D and Quantum Black architecture documentation and available infrastructure configuration.	Nam, Chris, Rick, and Dave Garber; before internal kickoff
Consolidate questions, gaps, dependencies, and architecture improvement opportunities.	LTI delivery team; for kickoff discussion
Hold the internal kickoff to align responsibilities, expectations, initial priorities, and questions for GE.	Monday, October 5, 2026 11:00 a.m. ET
Hold the GE kickoff to confirm scope, delivery priorities, dependencies, and next actions.	Tuesday, October 6, 2026 11:00 a.m. ET; invite sent by Russo

Action	Participants / Timing
Identify additional former DT4D stakeholders and coordinate involvement with Jim Tallman's engineering team.	Nam, Rick, Chris